

M13.8 Instruments — Batch 2

345 questions · Level 1: 150 · Level 2: 150 · Level 3: 45 · from the text you sent (LTT M13.08 pages 2–57)

Level 1 — recall

Classification

1. To which two subgroups can every instrument of the classical cockpit be assigned
 - a. Flight monitoring and engine monitoring
 - b. Airframe monitoring and aircraft monitoring
 - c. Flight monitoring and aircraft monitoring
2. A student is asked to sort every instrument of a classical cockpit. How many subgroups does the classification use?
 - a. Two subgroups
 - b. Three subgroups
 - c. Four subgroups
3. To which group does the artificial horizon indicator ADI belong
 - a. Flight monitoring
 - b. Airframe monitoring
 - c. Aircraft monitoring
4. To which group does the machmeter belong
 - a. Engine monitoring
 - b. Aircraft monitoring
 - c. Flight monitoring
5. To which group does an RPM indicator belong
 - a. Flight monitoring
 - b. Standby devices
 - c. Aircraft monitoring
6. To which group does the magnetic compass belong
 - a. Engine monitoring
 - b. Flight monitoring
 - c. Aircraft monitoring
7. Which of these instruments belongs to the flight monitoring group
 - a. Flap position indicator
 - b. Fuel quantity indicator
 - c. Turn and slip indicator
8. Which combination correctly identifies two devices that both belong to aircraft monitoring?
 - a. Position indicator and machmeter
 - b. Variometer and RPM indicator
 - c. Position indicator and RPM indicator
9. Which devices are the airframe monitoring devices of a classical cockpit
 - a. Position indicators and pressure indicators
 - b. Pressure indicators and fuel indicators
 - c. Position indicators and RPM indicators
10. Which devices are the engine monitoring devices of a classical cockpit
 - a. Position, exhaust gas temperature and fuel indicators
 - b. RPM, exhaust gas temperature and fuel indicators
 - c. RPM, exhaust gas temperature and position indicators
11. What does the abbreviation EIS stand for
 - a. Engine Indicating System
 - b. Electronic Instrument System
 - c. Electronic Indicating System
12. What does the abbreviation EFIS stand for
 - a. Electronic Flight Indicating System
 - b. Electronic Flight Information Source
 - c. Electronic Flight Instrument System

13. Which instruments are assigned to the EFIS in the modern cockpit
- The engine monitoring instruments
 - The flight monitoring instruments
 - The airframe monitoring instruments
14. What does the abbreviation EICAS stand for
- Engine Indicating and Centralized Alerting System
 - Electronic Indicating and Crew Alerting System
 - Engine Indicating and Crew Alerting System
15. What does the abbreviation ECAM stand for
- Electronic Centralized Aircraft Monitoring
 - Electronic Centralized Airframe Monitoring
 - Engine Centralized Aircraft Monitoring
16. What does the abbreviation CMS stand for
- Centralized Monitoring System
 - Central Management System
 - Central Maintenance System
17. Which devices make up the standby set of a modern cockpit
- Altimeter, airspeed indicator, machmeter and compass
 - Altimeter, airspeed indicator, horizon and compass
 - Altimeter, variometer, horizon and compass
18. How many standby devices are provided in the modern cockpit
- Four
 - Five
 - Three
19. A change in which of these values causes an instrument to operate
- Cabin humidity
 - Earth's magnetism
 - Solar radiation
20. A change in which combination of values causes instruments to operate
- Air pressure, acceleration and engine thrust
 - Air pressure, temperature and acceleration
 - Air pressure, temperature and humidity
21. Which gyro-related value is one of the values whose change causes an instrument to operate
- Gyro bearing friction
 - Gyro stability
 - Gyro rotor mass

Cockpit displays and the Basic T

22. What is provided to each pilot for flight attitude and flight monitoring
- His own set of engine and systems monitoring instruments
 - One set of flight attitude and flight monitoring instruments shared by both pilots
 - His own set of flight attitude and flight monitoring instruments
23. What does the Basic T set of flight monitoring instruments tell the pilot about
- Aircraft airspeed, altitude and fuel quantity
 - Aircraft attitude, heading and engine thrust
 - Aircraft attitude, airspeed and altitude
24. Which instrument of the Basic Six indicates the rate of climb or dive
- The turn and bank indicator
 - The directional gyro
 - The variometer
25. Which unit is used by the airspeed indicator of the Basic Six
- Feet
 - Knots
 - Degrees

26. An altimeter of the Basic Six set is being checked on the bench. Which unit should its scale be calibrated in?
- Knots
 - Metres
 - Feet
27. Which instrument of the Basic Six provides the compass function
- The artificial horizon
 - The directional gyro
 - The variometer
28. What does the artificial horizon of the Basic Six indicate
- Magnetic heading
 - Aircraft attitude
 - Rate of climb
29. How many essential instruments make up the set used for flying in instrument meteorological conditions
- Four
 - Six
 - Eight
30. A panel layout is described as Basic T. Which four instruments form that T?
- Airspeed indicator, attitude indicator, altimeter and compass
 - Attitude indicator, altimeter, compass and turn coordinator
 - Airspeed indicator, attitude indicator, altimeter and vertical speed indicator
31. Where is the attitude indicator located in the Basic T configuration
- On the left at the top
 - In the middle at the top
 - In the middle at the bottom
32. Which instrument is located to the right of the attitude indicator in the Basic T
- The airspeed indicator
 - The altimeter
 - The compass
33. Which instrument is located to the left of the attitude indicator in the Basic T
- The airspeed indicator
 - The altimeter
 - The vertical speed indicator
34. A panel is being rebuilt to the Basic T standard. Which instrument belongs directly below the attitude indicator?
- The altimeter
 - The compass
 - The airspeed indicator
35. Where is the magnetic standby compass often found
- Between the windscreens
 - Below the airspeed indicator
 - Beside the altimeter
36. A Basic T panel is being checked for correct arrangement. Which instruments form its upper row?
- Radio compass, directional indicator and altimeter
 - Airspeed, artificial horizon and vertical speed
 - Airspeed, artificial horizon and altimeter
37. Which instruments form the lower row of the Basic T
- Radio compass, directional indicator and vertical speed
 - Radio compass, directional indicator and altimeter
 - Airspeed, directional indicator and vertical speed
38. A flight instrument is listed in the wiring manual as the ADI. What does this abbreviation stand for?
- Attitude Deviation Indicator
 - Altitude Director Indicator
 - Attitude Director Indicator
39. What does an ADI display
- The artificial horizon and the indication from the radio altimeter
 - The compass and the indications from the radio navigation systems
 - The artificial horizon and the indication from the flight director system

40. What does HSI stand for
 - a. Horizontal Stabiliser Indicator
 - b. Horizontal Steering Indicator
 - c. Horizontal Situation Indicator
41. What is the name of the display each pilot has in an aircraft with modern displays
 - a. Primary Function Display
 - b. Primary Flight Display
 - c. Primary Flight Director
42. Which further Basic T parameter does the PFD show in addition to airspeed, artificial horizon, altitude and compass
 - a. Static air temperature
 - b. Vertical speed
 - c. Mach number
43. In the classic instrument arrangement, which indicator is placed below the ADI
 - a. The altitude indicator
 - b. The HSI
 - c. The airspeed indicator
44. Where is lateral deviation presented on the primary flight display
 - a. Along the left side of the display
 - b. Along the bottom of the display
 - c. Along the right side of the display
45. Where is vertical deviation presented on the primary flight display
 - a. Along the bottom of the display
 - b. Along the left side of the display
 - c. Along the right side of the display
46. What has happened to instrumentation in recent years
 - a. It has been greatly simplified and centralised
 - b. It has been greatly expanded and decentralised
 - c. It has been greatly simplified and duplicated
47. Which input methods follow the integrated central control units in the trend of control element integration
 - a. Touch input, viewing direction control and voice command
 - b. Touch input, pedal input and keyboard command
 - c. Switch input, viewing direction control and stick command
48. On which display is navigation information integrated
 - a. On the SD
 - b. On the ND
 - c. On the E/WD

The atmosphere and ISA

49. What is the proportion of nitrogen in dry, clean air directly above the surface of the earth
 - a. 1%
 - b. 78%
 - c. 21%
50. What proportion of dry, clean air directly above the surface of the earth is oxygen?
 - a. 11%
 - b. 78%
 - c. 21%
51. Dry, clean air at the surface is made up of nitrogen, oxygen and a remaining one per cent. Which gases form that one per cent?
 - a. Noble gases such as helium, argon, neon, krypton and xenon
 - b. By-products such as water vapour, carbon dioxide and ozone
 - c. Reactive gases such as hydrogen, methane and carbon monoxide
52. Which is the lowest layer of the atmosphere, surrounding the surface of the earth
 - a. The mesosphere
 - b. The stratosphere
 - c. The troposphere

53. In which layer of the atmosphere do the numerous weather activities take place
- In the stratosphere
 - In the troposphere
 - In the thermosphere
54. A flight is planned across the polar region. How high does the troposphere reach there?
- About 16.8 km
 - About 11 km
 - About 8.5 km
55. An aircraft climbs through the tropopause. Which layer does it enter?
- The mesosphere
 - The stratosphere
 - The thermosphere
56. What is the name of the boundary between the troposphere and the stratosphere
- The mesopause
 - The tropopause
 - The stratopause
57. What is the approximate upper limit of the stratosphere
- About 11 km
 - About 50 km
 - About 80 km
58. Which layer of the atmosphere contains the ozone layer?
- The stratosphere
 - The troposphere
 - The thermosphere
59. Which is the outermost layer of the atmosphere
- The exosphere
 - The mesosphere
 - The thermosphere
60. An air data system is being described to a student. Which two measurements does flight monitoring require?
- Static pressure and air density
 - Dynamic pressure and air temperature
 - Static pressure and dynamic pressure
61. What is the other name for dynamic pressure
- Atmospheric pressure
 - Ram pressure
 - Barometric pressure
62. What is the other name for static pressure
- Total pressure
 - Ram pressure
 - Atmospheric pressure
63. Which flight monitoring data are calculated from static and dynamic pressure
- Barometric altitude, vertical speed, airspeed and mach number
 - Barometric altitude, vertical speed, airspeed and static air temperature
 - Barometric altitude, true airspeed, mach number and static air temperature
64. Which air density is defined at mean sea level in the international standard atmosphere
- 1.325 kg/m³
 - 1.225 kg/m³
 - 0.765 kg/m³
65. Which temperature and pressure are defined at mean sea level in the international standard atmosphere
- 5 °C and 1013 mbar
 - 15 °C and 1013 mbar
 - 15 °C and 1030 mbar
66. Which pressure does a mechanical air data instrument use to indicate barometric altitude
- Static pressure
 - Total pressure
 - Dynamic pressure

67. In which layer does the summit of Mount Everest lie
- The troposphere
 - The tropopause
 - The stratosphere
68. What is the standard atmosphere sea level pressure in the imperial system
- 29.92 inHg
 - 14.7 inHg
 - 29.92 mmHg
69. What is the standard atmosphere sea level density in the imperial system
- 0.002378 lb/ft³
 - 0.002378 slug/ft³
 - 0.02378 slug/ft³
70. What is the standard atmosphere sea level temperature in the imperial system
- 15 °F
 - 32 °F
 - 59 °F
71. Which humidity is assumed in the standard atmosphere
- 100 %
 - 0 %
 - 50 %
72. A performance chart is drawn in metric units. Which temperature gradient applies from mean sea level to the tropopause?
- − 0.65 °C per 1000 m
 - − 0.65 °C per 100 m
 - − 6.5 °C per 100 m
73. Which air temperature is given at the tropopause in the ICAO standard atmosphere
- − 30.5 °C
 - − 56.5 °C
 - − 43.5 °C
74. Which air pressure is given at the tropopause in the ICAO standard atmosphere
- 226.32 hPa
 - 307.42 hPa
 - 165.10 hPa
75. The ICAO standard atmosphere table gives the tropopause as 11000 m. Which altitude in feet corresponds to it?
- 29527 ft
 - 36089 ft
 - 42650 ft
76. Which air density is given at 11000 m in the ICAO standard atmosphere
- 0.3692 kg m^{−3}
 - 0.2665 kg m^{−3}
 - 0.4663 kg m^{−3}
77. A mercury barometer is read at standard sea level conditions. Which pressure does it show?
- 1013 mm Hg
 - 760 mm Hg
 - 29.92 mm Hg
78. What is the standard atmosphere sea level pressure expressed in pounds per square inch
- 1.033 psi
 - 14.696 psi
 - 29.921 psi
79. What is the altitude of the tropopause in the international standard atmosphere
- 11 km
 - 8 km
 - 20 km

Altitude, altimeter settings and RVSM

80. What is the name of the screen on which modern indicating systems carry all the air data information
- The PFD
 - The SD
 - The ND
81. What does HUD stand for
- Head Up Display
 - Head Unit Display
 - Heading Up Display
82. Which air data parameter is presented on the left-hand scale of the PFD
- Vertical speed
 - Airspeed
 - Altitude
83. Where is the reference level set on the altimeter
- On the altitude drum
 - On the vertical speed scale
 - On the pressure scale
84. What is elevation
- The altitude of the aircraft above the ground
 - The altitude of an object on the ground above mean sea level
 - The altitude of the aircraft above mean sea level
85. An airfield is listed with its height above mean sea level. Which setting code belongs to this height definition?
- QNE
 - QFF
 - QFE
86. What is height
- The altitude of the aircraft above mean sea level
 - The altitude of an object on the ground above mean sea level
 - The altitude of the aircraft above the ground
87. Which instrument normally measures height
- The air data computer
 - The low range radio altimeter
 - The barometric altimeter
88. Which setting code belongs to altitude
- QFE
 - QFF
 - QNH
89. What is set on the altimeter when QNH is used
- The current air pressure at the airfield
 - The standard pressure of 1013.2 hPa
 - The current air pressure at mean sea level
90. Which setting code belongs to the standard setting used for flight levels
- QNE
 - QNH
 - QFF
91. Which air pressure are all pilots required to set for cruise above the transition altitude
- 1113.2 hPa
 - 1030 hPa
 - 1013.2 hPa
92. An altimeter is set to the standard pressure in cruise. What is its indication called?
- Pressure altitude
 - True altitude
 - Barometric height
93. What is a flight level
- The height above the mean sea level pressure of the day
 - The height above the airfield pressure level
 - The height above the 1013.25 hPa pressure level

94. What is the minimum margin by which the transition level lies above a transition altitude of 5000 ft MSL
- 1000 ft
 - 2000 ft
 - 500 ft
95. An operator prepares its fleet for RVSM operations. What does this abbreviation mean?
- The doubling of vertical safety distances at higher flight altitudes
 - The halving of vertical safety distances at higher flight altitudes
 - The halving of lateral safety distances at higher flight altitudes
96. Which minimum vertical separation is determined by the semicircular rule for IFR traffic up to FL 285
- 2000 ft
 - 500 ft
 - 1000 ft
97. Which vertical separation was applied above FL 285 until 1997
- 1000 ft
 - 4000 ft
 - 2000 ft
98. Which altitude change corresponds to a pressure change of 1 hPa at mean sea level
- 120 ft
 - 300 ft
 - 30 ft
99. An aircraft is cruising at FL 360 when the static pressure changes by 1 hPa. Which altitude change does this represent?
- 120 ft
 - 30 ft
 - 60 ft
100. What is RVSM airspace
- Airspace above FL 285 with 500 ft vertical separation
 - Airspace above FL 290 with 2000 ft vertical separation
 - Airspace above FL 290 with 1000 ft vertical separation
101. Which specification must an aircraft fulfil to be approved for RVSM
- MTBF
 - MNPS
 - MMEL
102. Which item is listed as an RVSM-sensitive component of the aircraft
- The inertial reference unit
 - The radio altimeter
 - The autopilot in altitude hold mode
103. An aircraft is being equipped for flight in RVSM airspace. How many independent altimeters must it have?
- 1
 - 2
 - 3
104. RVSM was first driven by the traffic over the Atlantic. In which further regions has it since been introduced?
- Over Europe, Africa and Pacific routes
 - Over Europe, America and polar routes
 - Over Europe, America and Pacific routes
105. Who must receive additional mandatory training for the introduction of RVSM
- Pilots and air traffic controllers
 - Pilots and maintenance personnel
 - Maintenance personnel and ground handlers
106. Corrosion of the pitot-static sensors is covered by the RVSM training. Which replacement interval is given for them?
- Every 9,600 hours
 - Every 6,900 hours
 - Every 960 hours
107. What does SSEC stand for
- Static Sensor Error Correction
 - Static Source Error Correction
 - Static Source Error Computation

- 108.** For which previously issued navigation approval is the MNPS Approval granted
- a.** ADC (Air Data Computer)
 - b.** LRN (Long Range Navigation)
 - c.** SSEC (Static Source Error Correction)

Airspeed

- 109.** What does the abbreviation IAS stand for
- a.** Indicated airspeed
 - b.** Increased airspeed
 - c.** Inertial airspeed
- 110.** Which airspeed is displayed in aircraft equipped with air data computers and glass cockpits
- a.** The true airspeed
 - b.** The calibrated airspeed
 - c.** The equivalent airspeed
- 111.** In the worked example, how does the CAS compare with the IAS
- a.** It is 7 knots higher
 - b.** It is 15 knots lower
 - c.** It is 7 knots lower
- 112.** Which two pressures are supplied to the air data computer for the airspeed computation
- a.** Static pressure and reference pressure
 - b.** Total pressure and static pressure
 - c.** Total pressure and dynamic pressure
- 113.** An air data computer receives an indicated airspeed of 300 kts. Which calibrated airspeed results in the example?
- a.** 278 kts
 - b.** 307 kts
 - c.** 293 kts
- 114.** What is the compressibility error at 30,000 ft with a calibrated airspeed of 293 knots
- a.** 15 knots
 - b.** 7 knots
 - c.** 22 knots
- 115.** For which purpose is the true airspeed needed
- a.** For navigation purposes
 - b.** For design data purposes
 - c.** For engine trim purposes
- 116.** An aircraft flies at 30,000 ft with an equivalent airspeed of 278 knots. Which true airspeed applies in the example?
- a.** 440 knots
 - b.** 293 knots
 - c.** 470 knots

Barometers, capsules and instrument errors

- 117.** How is the tube of a liquid-based barometer arranged
- a.** Horizontal and hermetically sealed at both ends
 - b.** Vertical and hermetically sealed at the bottom
 - c.** Vertical and hermetically sealed at the top
- 118.** Which height does mercury reach in a mercury barometer under normal conditions
- a.** 560 mm
 - b.** 760 mm
 - c.** 1013 mm
- 119.** What forms the sensing element of an aneroid barometer
- a.** A small capsule made of thin metal
 - b.** A small chamber made of thin glass
 - c.** A small tube filled with thin fluid
- 120.** What is the residual pressure inside an aneroid capsule
- a.** About 5 mbar
 - b.** About 50 mbar
 - c.** About 1013 mbar

121. Which value in pascal corresponds to the residual pressure of 5 mbar inside an aneroid capsule
- 5000 Pa
 - 50 Pa
 - 500 Pa
122. How many capsules are used at most in the stack of an improved barometer
- Up to sixteen
 - Up to eight
 - Up to four
123. What is the function of a Bourdon tube
- It converts pressure into a proportional temperature signal
 - It converts pressure into a proportional electrical signal
 - It converts vibration into a proportional electrical signal
124. Which cross-sectional shape does the curved tube of a Bourdon transducer have
- A square shape
 - A round shape
 - An oval shape
125. Which device is usually used as the position sensor of a Bourdon tube
- A vibration pickup
 - A resolver winding
 - A synchro transmitter
126. Which temperature range must a pressure sensor flight instrument cover
- 40 °C to +55 °C
 - 70 °C to +50 °C
 - 50 °C to +70 °C
127. How many revolutions must the altimeter needle make between -1,000 feet and +50,000 feet
- 15 revolutions
 - 51 revolutions
 - 50 revolutions
128. Which indication is given by the variometer of a pressure instrument panel
- The Mach number
 - The barometric altitude
 - The vertical speed
129. An instrument has to be connected to the aircraft pitot and static system. How is the pressure information brought to it?
- Via wiring and pressure transducers with screw or quick release connectors
 - Via piping and rigid tubes with soldered or welded connectors
 - Via piping and flexible tubes with screw or quick release connectors
130. Which stroke do diaphragm capsules supply as the measurable value
- 0.4 to 0.5 mm
 - 4 to 5 mm
 - 14 to 15 mm

Pressure transducers

131. What is the function of a pressure transducer
- To convert measured pressures into mechanical needle movement
 - To convert electrical signals into measured pressures
 - To convert measured pressures into electrical signals
132. In which equipment are pressure transducers used
- In inertial reference units
 - In air data computers
 - In flight control computers
133. Which type of output signal does the latest generation of pressure transducers deliver
- A synchro signal
 - An analog signal
 - A digital signal

134. A modern pressure transducer needs a fixed reference for its measurement. Which reference value is used?
- A vacuum or 1013 mbar
 - An ambient pressure or 5 mbar
 - A vacuum or 760 mbar
135. Which output signal is produced by a pressure cell transducer with a resolver
- A digital signal
 - An analog signal
 - A frequency signal
136. A force compensated pressure cell is opened. Which two electrical conversion elements are found in it?
- A torsion bar and an E-pickup
 - An oscillator and a resonator
 - A resolver and an E-pickup
137. Which elements perform the electrical conversion in a vibration pressure transducer
- A resolver and a torsion bar
 - A resistive bridge and an E-pickup
 - An oscillator and a resonator
138. Which element performs the electrical conversion in a piezoresistive pressure transducer
- A resolver output
 - A resistive bridge
 - A torsion bar link
139. Several electronic pressure sensor principles exist. Which one is described as the most commonly used?
- The vibrating transducer
 - The piezoresistive transducer
 - The force compensated cell
140. Which three parts make up the vibrating transducer
- A base plate, a center body and a torsion bar
 - A diaphragm, a center body and an outer case
 - A base plate, a center body and an outer case
141. What is normally inside the reference pressure chamber of a vibrating transducer
- Usually pitot air
 - Usually ambient air
 - Usually a vacuum
142. From which material is the micromechanical pressure transducer chamber made
- Pyrex material
 - Metal material
 - Silicon material
143. What is the size of the micromechanical pressure transducer structure
- Only a few centimetres
 - Only a few millimetres
 - Only a few micrometres
144. Which material is used together with the silicon in the micromechanical transducer
- Glass of the Pyrex type
 - Metal of the capsule type
 - Ceramic of the sintered type
145. What is denoted by UIN and UOUT in the piezoresistive transducer
- The reference voltage and the bridge voltage
 - The supply voltage and the measuring voltage
 - The measuring voltage and the supply voltage
146. In which circuit are the strain gauge resistances connected
- In a resolver winding
 - In a series chain
 - In a Wheatstone bridge
147. How many strain gauges are normally fitted on the diaphragm of a piezoresistive transducer
- Two resistors
 - Eight resistors
 - Four resistors

148. The bridge of a piezoresistive transducer is balanced. Which voltage then appears across the front resistors and across the back resistors

- a.** 8 volts on each side
- b.** 10 volts on each side
- c.** 5 volts on each side

149. In the worked example the voltage U_1 rises to 8 volts when pressure is applied. Which value does U_2 take?

- a.** 2 volts
- b.** 5 volts
- c.** 6 volts

150. Which supply voltage is applied to the strain gauge bridge in the example

- a.** 10 volts
- b.** 8 volts
- c.** 5 volts

Level 2 — understand and explain

Classification

151. An exhaust gas temperature indicator is fitted in a classical cockpit. Which group and subdivision does it belong to?
- Aircraft monitoring, as an airframe monitoring device
 - Flight monitoring, as a pressure monitoring device
 - Aircraft monitoring, as an engine monitoring device
152. Into which two kinds of devices is aircraft monitoring divided
- Flight monitoring devices and engine monitoring devices
 - Airframe monitoring devices and standby devices
 - Airframe monitoring devices and engine monitoring devices
153. A pressure indicator for a hydraulic system is fitted in the cockpit. Which statement best describes its classification?
- It is a flight monitoring instrument within the classical group
 - It is an airframe monitoring device within aircraft monitoring
 - It is an engine monitoring device within aircraft monitoring
154. The classical cockpit of an aircraft is replaced by a glass cockpit. Which display system takes over the altimeter and airspeed indications?
- The CMS, which carries the flight monitoring instruments
 - The EFIS, which carries the flight monitoring instruments
 - The EICAS, which carries the flight monitoring instruments
155. Why do the aircraft monitoring systems of the modern cockpit carry different names such as EICAS or ECAM
- Because the name depends on the flight phase
 - Because the name depends on the display size
 - Because the name depends on the aircraft type
156. Which task do EICAS and ECAM perform in the modern cockpit
- They perform flight monitoring
 - They perform aircraft monitoring
 - They perform maintenance data recording
157. An engine RPM value must be presented to the crew of a glass cockpit aircraft. Which system group presents it?
- The flight monitoring system, the EFIS
 - The maintenance monitoring system, the CMS
 - The aircraft monitoring system, EICAS or ECAM
158. Why is the CMS counted as part of aircraft monitoring
- Because it delivers information to the flight crew
 - Because it delivers information to the maintenance personnel
 - Because it delivers information to the standby instruments
159. Which combination correctly pairs the monitoring task with the system of a modern glass cockpit?
- Flight monitoring with ECAM, aircraft monitoring with EFIS
 - Flight monitoring with EFIS, aircraft monitoring with ECAM
 - Flight monitoring with EIS, aircraft monitoring with CMS
160. A student claims the machmeter is part of the standby set. Which statement correctly names the standby devices?
- They are the altimeter, airspeed indicator, horizon and compass
 - They are the altimeter, machmeter, horizon and compass
 - They are the airspeed indicator, machmeter, horizon and compass
161. A fuel pressure indication changes during flight. Which value is the instrument responding to?
- Pressure of air
 - Pressure of fluids
 - Earth's magnetism
162. A compass reacts to a change in which of these values?
- Gyro stability
 - Earth's magnetism
 - Fluid pressure
163. An altimeter and an airspeed indicator both respond to a change in which value?
- Air pressure
 - Fluid pressure
 - Acceleration

Cockpit displays and the Basic T

164. The captain's flight attitude and monitoring instruments fail completely. Which statement applies to the first officer?
- He has his own set of engine and systems monitoring instruments
 - He has his own complete set of flight attitude and monitoring instruments
 - He shares one central set of flight attitude and monitoring instruments
165. Light trainers and heavy bombers of one air force were fitted with the same six instruments in the same panel positions. Which advantage does this bring?
- A pilot trained on one type can quickly get used to flying another type blind
 - A pilot trained on one type needs no attitude instrument on another type
 - A pilot trained on one type can fly another type without an airspeed indicator
166. A pilot reads his magnetic heading and a VOR course deviation from one round instrument. Which indicator is he using?
- The ADI
 - The variometer
 - The HSI
167. Which combination correctly matches the two indicators with the information they present?
- ADI: attitude with radio navigation; HSI: compass with flight director
 - ADI: attitude with flight director; HSI: compass with radio navigation
 - ADI: compass with radio navigation; HSI: attitude with flight director
168. A glass cockpit aircraft has no separate round flight instruments. Which statement describes the arrangement of the flight parameters?
- They are laid out on the PFD in a configuration that replaces the Basic T
 - They are laid out on the PFD and correspond to the Basic T configuration
 - They are laid out on the navigation display in Basic T configuration
169. On a glass cockpit primary flight display, which information occupies the central area?
- Lateral deviation
 - Vertical deviation
 - Attitude and guidance
170. Switches, sticks, pedals, keys and knobs were once fitted as individual operating devices. Into what does the trend in control elements integrate them?
- Into integrated central control units
 - Into dedicated standby instruments
 - Into separate air data calculators
171. On which displays is attitude information integrated in the display integration trend
- On the PFD and the HUD
 - On the E/WD and the SD
 - On the ND and the MFD
172. Engine parameters and system synoptics are being grouped onto the centre displays. Which combination applies?
- Engine on the E/WD and systems on the SD
 - Engine on the SD and systems on the E/WD
 - Engine on the ND and systems on the E/WD

The atmosphere and ISA

173. An aircraft cruises at the same pressure altitude first over the equator and then over the pole. Where is the troposphere thicker?
- Over the equator, where it is about 16.8 km thick
 - Over the pole, where it is about 16.8 km thick
 - Over the equator, where it is about 8.5 km thick
174. How does the temperature behave with increasing height in the troposphere
- It decreases by 0.65 °C every 100 metres
 - It increases by 0.65 °C every 100 metres
 - It decreases by 6.5 °C every 100 metres
175. Which is the layer above the mesopause in which the temperature increases again with height
- The stratosphere
 - The mesosphere
 - The thermosphere

176. How do static pressure, temperature and density behave as altitude increases
- Only temperature becomes lower
 - All three become higher
 - All three become lower
177. True airspeed and static air temperature are to be indicated as well. Which value must be applied as a corrective?
- Density
 - Humidity
 - Temperature
178. A pneumatic instrument is calibrated in the workshop. Which values form the basis of that calibration?
- The tropopause values of the standard atmosphere
 - The standard atmosphere values at mean sea level
 - The actual airfield values at the time of installation
179. A uniform reference value was needed for aviation. Which body introduced the international standard atmosphere?
- IATA
 - EASA
 - ICAO
180. An aircraft climbs from sea level towards the tropopause. Which trend applies to the surrounding air?
- Static pressure, temperature and density all decrease
 - Static pressure, temperature and density all increase
 - Static pressure and density decrease while temperature increases
181. Why has the international standard atmosphere been established
- To set the pressure scale of the altimeter before every flight
 - To compare flight performance under greatly differing conditions
 - To predict the weather conditions along a planned route
182. Density in the standard atmosphere is given as a ratio of the sea level value. Which value does that ratio have at sea level?
- 0 %
 - 100 %
 - 50 %

Altitude, altimeter settings and RVSM

183. Where can the conversion from pressure to indication take place
- In the instrument itself or separately in air data calculators
 - In the instrument itself or separately in the transponder
 - In the pitot probe itself or separately in the instrument
184. An aircraft has separate air data calculators feeding electrical instruments. What does the calculator do with the pressure?
- It converts the pressure mechanically into electricity for the indicators
 - It converts the pressure into a mechanical drive for the indicators
 - It converts the electricity from the indicators back into pressure
185. Two aircraft types show the same air data but work differently inside. On what does the way the instruments function depend?
- On the operator and the altitude monitoring programme
 - On the manufacturer and the state of technological development
 - On the manufacturer and the reference pressure selected
186. How does the HUD present the PFD data
- So that the pilot can see it when he looks down at the centre pedestal
 - So that the pilot can see it when he looks through the windshield
 - So that the pilot can see it only when the autopilot is engaged
187. A pilot keeps looking straight ahead through the windshield during the approach and still reads his speed and altitude. Which display is he using?
- The head up display
 - The multi function display
 - The primary flight display
188. A barometric altimeter is set to QFE while the aircraft stands on the airfield. What does it indicate?
- Zero altitude
 - Current flight level
 - Airfield elevation

- 189.** What does a barometric altimeter set to QFE indicate in flight
- The altitude above mean sea level, as long as the air pressure does not change
 - The height above the airfield, as long as the air pressure does not change
 - The flight level, as long as the air pressure does not change
- 190.** An aircraft stands on an airfield with QNH set on its altimeter. What does the altimeter show?
- Airfield elevation
 - Zero altitude
 - Transition level
- 191.** A crew selects QNH on both altimeters. In which phases of flight is this setting used?
- During landing and take-off, above transition altitude and above transition level
 - During cruise, above transition altitude and above transition level
 - During landing and take-off, up to transition altitude and below transition level
- 192.** Above which altitude do all pilots set the standard pressure in cruise flight
- 5000 ft above mean sea level
 - 10000 ft above mean sea level
 - 2000 ft above mean sea level
- 193.** All aircraft in cruise use the same standard pressure setting. Which purpose does this serve?
- It ensures correct indication of the airfield elevation
 - It ensures a correct height above the airfield
 - It ensures safe vertical separation
- 194.** What is the transition altitude
- After take-off, the height at which the altimeter is set from QNH to STD
 - Prior to landing, the height at which the altimeter is set from QNH to STD
 - After take-off, the height at which the altimeter is set from STD to QNH
- 195.** What is the transition level
- Prior to landing, the last usable flight level at which the altimeter is set from QNH to STD
 - Prior to landing, the last usable flight level at which the altimeter is set from STD to QNH
 - After take-off, the last usable flight level at which the altimeter is set from STD to QNH
- 196.** Which value on the altimeter static pressure scale corresponds to the ISA sea level reference
- 1030 hPa
 - 1013.25 hPa
 - 935 hPa
- 197.** An aircraft cruises with the standard setting selected and the flight level display shows FL 50. Which pressure surface is its reference?
- The 1013.25 hPa pressure surface
 - The mean sea level pressure of the day
 - The airfield pressure surface
- 198.** What is the purpose of reducing the vertical separation minimum
- To accommodate more aircraft in the same airspace
 - To reduce the workload of the air data computers
 - To allow aircraft to climb above the tropopause
- 199.** Why was the separation above FL 285 formerly increased
- Lack of measurement precision and reduced altimeter sensitivity at high altitude
 - Lack of radar coverage and reduced transponder range at high altitude
 - Lack of autopilot authority and reduced control response at high altitude
- 200.** What does static source error depend on
- The fixed location of the static pressure inlet
 - The fixed location of the air data computer
 - The selected reference pressure on the altimeter
- 201.** Work is planned around the static ports of an RVSM aircraft. Which condition must the structure and surroundings of the pitot-static sensors be left in?
- Clean and clear
 - Sealed and painted
 - Smoothed with filler

- 202.** What is the function of the warning device required for flight in RVSM airspace
- It indicates when the aircraft enters RVSM airspace
 - It indicates when the aircraft leaves the predetermined height
 - It indicates when the altimeter setting differs from 1013.2 hPa
- 203.** Which autopilot capability is required for flight in RVSM airspace
- It must be able to maintain the altitude
 - It must be able to maintain the airspeed
 - It must be able to maintain the heading
- 204.** Which transponder requirement applies to an aircraft flying in RVSM airspace?
- It must transmit altitude and be connected to the altitude measurement unit
 - It must transmit heading and be connected to the altitude measurement unit
 - It must transmit altitude and be connected to the radio altimeter
- 205.** Which operational advantage is gained by an RVSM approved aircraft
- It can use a reduced number of flight levels to save maintenance costs
 - It can use the increased number of flight levels to save time and fuel
 - It can use the increased number of flight levels to fly above the tropopause
- 206.** When RVSM is fully implemented, between which flight levels will the RVSM and MNPS requirements apply?
- From FL 270 to FL 430
 - From FL 290 to FL 410
 - From FL 290 to FL 420
- 207.** An aircraft is to fly between FL 290 and FL 420 once RVSM is fully implemented. Which authorization must it hold?
- Both MNPS and RVSM authorization
 - RVSM authorization only
 - MNPS authorization only
- 208.** A fleet of six aircraft is to be cleared for RVSM. Who must apply for the Airworthiness Approval and for what?
- The aircraft manufacturer, for each individual aircraft
 - The aircraft operator, for the whole fleet at once
 - The aircraft operator, for each individual aircraft
- 209.** What must each aircraft undergo before RVSM Airworthiness Approval
- Replacement of the air data computers and the altitude monitoring programme
 - Modification according to required service bulletins and a full static system overhaul
 - Modification according to required service bulletins and the altitude monitoring programme
- 210.** The RVSM training covers surface waviness in the critical area. Which causes of that waviness are named?
- Fasteners, oxygen discharge discs and paint drops
 - Fasteners, static dischargers and paint drops
 - Rivets, oxygen discharge discs and corrosion pits
- 211.** Which further subject of the mandatory RVSM training concerns the air data computer itself?
- Its cooling airflow
 - Its software loading procedure
 - Its part numbers
- 212.** Which approval makes the aircraft eligible for the RVSM test flight
- The RVSM Operational Approval
 - The RVSM Airworthiness Approval
 - The MNPS Approval
- 213.** The altitude monitoring programme and the test flight have both been completed successfully. Which approval does the operator now receive?
- The MNPS Approval
 - The RVSM Airworthiness Approval
 - The RVSM Operational Approval
- 214.** Which documents are issued in revised form once the RVSM conditions are fulfilled
- The operation and maintenance manuals
 - The operation and training manuals
 - The maintenance and weight and balance manuals

Airspeed

- 215.** An airspeed indicator with bellows is connected to the pitot and static systems. Which pressure does its indication directly follow?
- The static pressure as the difference between ambient and total pressure
 - The dynamic pressure as the difference between total and static pressure
 - The total pressure as the sum of dynamic and ambient pressure
- 216.** Under which conditions does an airspeed indicator with bellows show the real speed of the aircraft
- At mean sea level with compressibility corrected
 - At mean sea level with standard conditions
 - At cruise altitude with standard conditions
- 217.** An air data computer converts the indicated airspeed into the calibrated airspeed. Which two errors does it remove?
- The error from the indicator and the error from the static pressure source
 - The error from the indicator and the error from air compressibility
 - The error from the pitot source and the error from air density
- 218.** A mechanical standby airspeed indicator with bellows is fed with total and static pressure. Which airspeed does it display?
- The indicated airspeed
 - The equivalent airspeed
 - The calibrated airspeed
- 219.** Which unit performs the correction of instrument error and static source error
- The air data computer
 - The airspeed indicator
 - The pitot probe
- 220.** Under which condition is the speed formula valid
- Only when the medium used is incompressible
 - Only when the medium used is above the tropopause
 - Only when the medium used is at standard temperature
- 221.** An aircraft accelerates to a high speed and the total pressure rises more than the speed formula predicts. What produces this additional pressure?
- Air expands at higher speeds and generates a total pressure decrease
 - Air compresses at higher speeds and generates a total pressure increase
 - Air compresses at lower speeds and generates a static pressure increase
- 222.** How does the compressibility error change with airspeed and altitude
- The higher the airspeed and altitude, the smaller the error
 - The lower the airspeed and the higher the altitude, the larger the error
 - The higher the airspeed and altitude, the larger the error
- 223.** What is the equivalent airspeed
- The true airspeed after the density calculation has been removed
 - The indicated airspeed after the instrument error has been removed
 - The calibrated airspeed after the compressibility error has been removed
- 224.** Why is the equivalent airspeed not shown to the crew of a commercial airplane
- It is only important for design data and further calculations
 - It is only available when the air data computer is in test
 - It is only valid below the tropopause in standard conditions
- 225.** A navigation computation needs the speed of the aircraft through the surrounding air mass. Which airspeed is this?
- The equivalent airspeed
 - The calibrated airspeed
 - The true airspeed
- 226.** In which order are the airspeeds derived from the pressure inputs
- CAS, IAS, TAS, EAS
 - IAS, EAS, CAS, TAS
 - IAS, CAS, EAS, TAS
- 227.** Which calculation is made between EAS and TAS
- A static source correction
 - A density calculation
 - A compressibility correction
- 228.** Which correction is applied between CAS and EAS
- Correction of compressibility error
 - Correction of static source error
 - Correction of instrument error

229. The compressibility error at a given flight condition is stated as a number of knots. What is done with this value to obtain the equivalent airspeed?
- a. It is added to the calibrated airspeed
 - b. It is subtracted from the true airspeed
 - c. It is subtracted from the calibrated airspeed

Barometers, capsules and instrument errors

230. What creates the negative pressure at the top of a liquid barometer tube
- a. The expansion of the glass tube with temperature
 - b. The ambient air pressure acting on the container
 - c. The fluid flowing out of the tube due to its own weight
231. The mercury column of a barometer stops moving at a certain height. Which pressure balances the outflow of the fluid?
- a. The negative pressure counteracting the ambient air pressure
 - b. The ambient air pressure counteracting the outflow of the fluid
 - c. The temperature expansion of the fluid in the container
232. Which effect must be taken into account when reading a mercury barometer
- a. Both the mercury and the glass tube contract with increasing temperature
 - b. Only the mercury expands while the glass tube stays unchanged
 - c. Both the mercury and the glass tube expand with increasing temperature
233. A mercury barometer is read in a workshop that is warmer than standard. What must be done to obtain an exact result?
- a. The reading must be adjusted to standard conditions
 - b. The reading must be adjusted to the residual capsule pressure
 - c. The reading must be adjusted to the local field elevation
234. The capsule of an aneroid barometer holds a residual pressure of about 5 mbar. Which effect does this residual pressure compensate?
- a. It compensates for changes in the elasticity of the metal caused by temperature
 - b. It compensates for changes in the friction of the mechanism caused by wear
 - c. It compensates for changes in the density of the air caused by altitude
235. A barograph is built with a stack of capsules instead of a single one. Why is this done?
- a. To increase the residual pressure of the capsules
 - b. To increase the temperature range of the instrument
 - c. To increase the reliability of measurements
236. How does an aneroid capsule react to increasing air pressure
- a. It is unchanged
 - b. It is compressed
 - c. It is expanded
237. An aircraft climbs and the static pressure around the aneroid capsule falls. How does the capsule react?
- a. It vibrates
 - b. It contracts
 - c. It expands
238. The aneroid capsule of a barometer changes its shape with pressure. How does this movement reach the indication?
- a. It is transferred hydraulically to an indicating needle
 - b. It is transferred electrically to an indicating needle
 - c. It is transferred mechanically to an indicating needle
239. Which pressure is applied to the outside of the capsule in an aneroid altimeter
- a. The static pressure
 - b. The dynamic pressure
 - c. The total pressure
240. In a mercury barometer, where does the ambient pressure act and where is the negative pressure found?
- a. Ambient pressure acts above the column, negative pressure is in the container
 - b. Ambient pressure acts on the container, negative pressure is above the column
 - c. Ambient pressure acts on both ends, negative pressure is inside the mercury
241. Which additional provision, beyond the pressure capsule and the indicating linkage, is built into an aneroid altimeter
- a. A compensation for the effect of humidity
 - b. A compensation for the effect of acceleration
 - c. A compensation for the effect of temperature

- 242.** A Bourdon tube transducer is opened for inspection. Which two parts carry out the pressure conversion?
- The straight tube and a position sensor
 - The curved tube and a drive coil
 - The curved tube and a position sensor
- 243.** A Bourdon tube transducer is fitted with a synchro transmitter. Which quantity does this sensor convert into an electrical signal?
- The temperature of the tube wall
 - The pressure inside the tube
 - The position of the tube tip
- 244.** How is an applied pressure distributed inside a Bourdon tube
- Only at the oval section of the tube
 - Equally throughout the whole tube
 - Only at the sealed tube tip
- 245.** What does the applied pressure try to do to the cross section of a Bourdon tube
- Change it from oval to circular
 - Change it from circular to oval
 - Change it from oval to rectangular
- 246.** Pressure is applied to a Bourdon tube and its oval section is pushed towards a circular one. How does the tube react?
- A further curving of the tube that moves the tip to the left
 - A straightening of the tube that moves the tip to the right
 - A lengthening of the tube that moves the tip downwards
- 247.** What is the amount of tube tip movement proportional to
- The tube temperature
 - The signal frequency
 - The applied pressure
- 248.** Which property is required from a pressure sensor flight instrument
- Sensitivity to acceleration
 - Insensitivity to dynamic pressure
 - Insensitivity to acceleration
- 249.** Why is there little leeway for manufacturers of pressure sensor instruments
- Most of the requirements are defined by the airline operator
 - Most of the requirements are defined and required by law
 - Most of the requirements are defined by the manufacturers
- 250.** Which instruments still use pressure sensors today
- Only back-up instruments
 - Only primary flight displays
 - Only engine indications
- 251.** Several thin metal diaphragm capsules are stacked into a single unit in a pressure sensor instrument. What is this assembly called?
- A bellows
 - A torsion bar
 - A resonator
- 252.** Which group of cockpit instruments is operated from pitot and static pressure?
- Airspeed indicator, altimeter, machmeter and attitude indicator
 - Airspeed indicator, altimeter, machmeter and variometer
 - Airspeed indicator, altimeter, variometer and radio altimeter
- 253.** A diaphragm capsule delivers only a stroke of 4 to 5 mm with a small force. What must the transfer mechanism do with it?
- Convert it into an amplified stroke of the capsule
 - Convert it into an amplified rotation of the needle
 - Convert it into a damped rotation of the needle
- 254.** Which effect does friction have in the transfer mechanism of an aneroid instrument
- It raises the capsule hysteresis
 - It falsifies the measured result
 - It damps the capsule stroke
- 255.** How is friction in the indicator mechanism kept to a minimum
- With sealed bearings and grease filling of all moving parts
 - With suitable bearings and clean construction of all moving parts
 - With heavier levers and increased spring tension on all moving parts

- 256.** In the altimeter the fight against friction is additionally supported by one further measure. Which measure is this?
- Balancing of the needle
 - Damping of the capsule
 - Vibration of the instrument
- 257.** Play in the gearing between the pinion and the tooth segment would falsify the indication. How is this play taken out?
- By increased bearing clearance
 - By spring tension on one side
 - By spring tension on both sides
- 258.** What is meant by the hysteresis of a membrane capsule
- The curve with increasing pressure differs from the curve with decreasing pressure
 - The curve with increasing pressure differs from the curve with increasing temperature
 - The stroke lags behind the pressure when a constant pressure acts for a long time
- 259.** What causes the elastic aftereffect of a capsule
- A constant pressure acting on the capsule over a longer period of time
 - A temperature change acting on the capsule mounting
 - A rapidly changing pressure acting on the capsule during a climb
- 260.** What is the result of the slow yielding of a capsule under constant pressure
- A change of the residual pressure
 - A permanent offset of the needle
 - A time lag in the stroke
- 261.** Transfer mechanism, hysteresis, elastic aftereffect and temperature are listed together. Which instruments do these error influences affect?
- Only digital air data computers
 - All aneroid type instruments
 - Only metric scale altimeters
- 262.** Which part provides the temperature compensation in the mechanical altimeter mechanism
- A bimetal element
 - A spring tensioned pinion
 - A second diaphragm capsule
- 263.** The hysteresis and the elastic aftereffect of a capsule are shown as characteristic curves. Which quantity of the capsule is plotted against the applied pressure?
- The diaphragm stroke
 - The needle friction
 - The residual capsule pressure
- 264.** Which part of the altimeter mechanism drives the indicating needle directly
- The bimetal rod on the bellows mounting
 - The pinion on the pointer axis
 - The steel pins in the capsule stack

Pressure transducers

- 265.** Pressure transducers have converted the measured pressures into electrical signals. What happens with these signals next?
- They are sent to the probe heater and transformed into the required current
 - They are sent to the computer and transformed into the required output signals
 - They are sent to the indicator and transformed into the required needle deflection
- 266.** How did the first pressure transducers work
- With metal pressure capsules, electromotively according to the follow-up principle
 - With metal pressure capsules, mechanically according to the resonance principle
 - With semiconductive pressure chambers, electrically according to the bridge principle
- 267.** From which material are the pressure chambers of the latest generation of transducers made
- Semiconductive material
 - Thin metallic material
 - Pyrex glass material
- 268.** What is stored in the integrated memory of a latest generation pressure transducer
- The reference voltage
 - The error characteristics
 - The measured pressures

- 269.** In the pressure cell type of transducer the pressure acts on one side of a metal diaphragm. What is on the other side?
- A vacuum space
 - A voltage supply
 - A pitot supply
- 270.** Which characteristic applies to the simple pressure cell transducer
- It is sensitive to pressure changes and mechanically complex
 - It is sensitive to pressure changes and electronically complex
 - It is insensitive to pressure changes and mechanically complex
- 271.** How are the instrument errors of a vibration pressure transducer handled
- They are compensated mechanically
 - They are compensated electrically
 - They are compensated by a torsion bar
- 272.** Which kind of complexity is stated for the vibration pressure transducer
- It is electronically complex
 - It is pneumatically complex
 - It is mechanically complex
- 273.** Which two transducer types deliver a digital output signal?
- The pressure cell and the piezoresistive transducer
 - The pressure cell and the force compensated pressure cell
 - The vibration and the piezoresistive transducer
- 274.** Which transducer types deliver an analog output signal?
- The pressure cell and the force compensated pressure cell
 - The pressure cell and the vibration pressure transducer
 - The force compensated cell and the piezoresistive transducer
- 275.** Which output does a vibrating transducer generate
- An output frequency which depends on the input pressure
 - An output frequency which depends on the reference pressure
 - An output voltage which depends on the input pressure
- 276.** Which function does the outer case of a vibrating transducer have
- It forms the outer wall of the measured pressure chamber
 - It forms the inner wall of the reference pressure chamber
 - It forms the outer wall of the reference pressure chamber
- 277.** Which part forms the inner wall of the reference chamber
- The center body part
 - The base plate part
 - The vibrating cylinder
- 278.** The cylinder of a vibrating transducer starts to vibrate at its natural frequency. What has caused this?
- A magnetic field in the pickup coil
 - A magnetic field in the drive coil
 - A pressure pulse in the reference chamber
- 279.** On what does the natural vibrating frequency of the cylinder depend
- On the body itself and the output signal frequency
 - On the drive coil current and the amplifier gain
 - On the body itself and the environmental conditions
- 280.** Which task does the pickup coil have in a vibrating transducer
- It measures the vibrating frequency of the cylinder
 - It seals the reference chamber against the measured pressure
 - It drives the cylinder at its natural frequency
- 281.** In the vibrating transducer the measured pressure enters at one end and a signal leaves through an amplifier. Which signal leaves the unit?
- A voltage
 - A resistance
 - A frequency
- 282.** Which physical behaviour works with the same forces as the vibrating cylinder
- The natural frequency of violin strings or tuning forks
 - The resonance of an aneroid capsule under constant pressure
 - The deflection of a Bourdon tube under internal pressure

- 283.** What happens when pressure acts on the silicon lid of a piezoresistive transducer
- The lid is dented and the resistance in its surface changes
 - The lid is dented and the capacitance in its surface changes
 - The lid vibrates and the frequency of its surface changes
- 284.** Which quantity is the measurement of pressure in a piezoresistive transducer
- The change in supply voltage
 - The change in reference pressure
 - The change in electrical resistance
- 285.** The bridge circuit of a micromechanical transducer is under pressure. How are the strain gauge resistances R1 and R2 strained?
- R1 and R2 both stretched
 - R1 compressed and R2 stretched
 - R1 stretched and R2 compressed
- 286.** What is the output voltage of a piezoresistive transducer proportional to
- The input pressure
 - The reference pressure
 - The supply frequency
- 287.** A piezoresistive transducer feeds an air data computer. What is done with its output voltage before the computer uses it?
- It is rectified
 - It is resolved
 - It is digitized
- 288.** The micromechanical transducer has two chambers separated by the diaphragm. Which chambers are these?
- A chamber for input pressure and a chamber for reference pressure
 - A chamber for total pressure and a chamber for reference pressure
 - A chamber for input pressure and a chamber for static pressure
- 289.** The characteristic of a piezoresistive pressure transducer is a straight rising line. What does this tell you about its output voltage?
- It rises in steps as the applied pressure rises
 - It rises in proportion to the applied pressure
 - It stays constant while the applied pressure rises
- 290.** How are the four strain gauges distributed on the base plate
- Three on the front side and one on the back side
 - Two on the front side and two on the back side
 - All four on the front side of the plate
- 291.** No input pressure is applied to a piezoresistive transducer. What is the condition of the base plate and the bridge?
- The plate is straight, all resistors are equal and the bridge is balanced
 - The plate is straight, the front resistors are lower and the bridge is balanced
 - The plate is bent, all resistors are equal and the bridge is balanced
- 292.** Which output voltage results when the bridge is balanced
- 0 volts
 - 10 volts
 - 5 volts
- 293.** What happens to the resistances when pressure bends the base plate
- Both the front and the back resistance increase
 - The front resistance decreases and the back resistance increases
 - The front resistance increases and the back resistance decreases
- 294.** What is the result of the resistance change when the base plate bends
- An imbalance in the supply which causes a drop in the input voltage
 - An imbalance in the bridge which causes an imbalance in the output
 - A balance in the bridge which causes a constant output
- 295.** Which two factors determine the resistance of a strain gauge wire?
- The type of material and the applied supply voltage
 - The type of material and the dimension of the wire
 - The bridge balance and the dimension of the wire
- 296.** How does the material constant of a strain gauge behave during bending
- It remains the same value
 - It rises with the bending
 - It falls with the bending

- 297.** How does the resistance of a strain gauge change with the length of its wire
- a. It decreases with the length
 - b. It is independent of the length
 - c. It increases with the length
- 298.** The four strain gauges of the bridge form two pairs on opposite faces of the plate. Which resistors form these pairs?
- a. R1 with R4 and R2 with R3
 - b. R1 with R3 and R2 with R4
 - c. R1 with R2 and R3 with R4
- 299.** How is the resistance of a strain gauge wire calculated
- a. Material constant times length divided by cross-sectional area
 - b. Length times cross-sectional area divided by material constant
 - c. Material constant times cross-sectional area divided by length
- 300.** A piezoresistive transducer is at rest with no pressure applied. What is the state of the strain gauge grid on the diaphragm?
- a. It is flat and unstrained
 - b. It is flat but compressed
 - c. It is bent and stretched

Level 3 — apply and reason

Classification

301. After landing, an engineer needs system information gathered during the flight rather than the indications the crew watched. Which system is intended for this and why?

- a. The EFIS, because it delivers the flight monitoring information
- b. The EIS, because it delivers all indications of the glass cockpit
- c. The CMS, because it delivers information to the maintenance personnel

302. Two glass cockpit aircraft of different types both display the same two monitoring groups. Which statement best describes the difference in naming?

- a. The flight monitoring system is the EFIS on both, while the aircraft monitoring system is named differently
- b. Both the flight and the aircraft monitoring systems are named differently on each type
- c. The aircraft monitoring system is the EFIS on both, while the flight monitoring system is named differently

303. The electronic displays of a glass cockpit are lost in flight. Which speed and attitude information remains available from the standby set?

- a. Airspeed from the standby airspeed indicator and heading from the standby horizon
- b. Mach number from the standby machmeter and attitude from the standby horizon
- c. Airspeed from the standby airspeed indicator and attitude from the standby horizon

The atmosphere and ISA

304. An aircraft climbs 2000 m inside the troposphere under standard conditions. Which temperature change results?

- a. A decrease of about 1.3 °C
- b. An increase of about 13 °C
- c. A decrease of about 13 °C

305. The standard atmosphere table gives −56.5 °C at 11000 m and again at 13000 m. What does this show about the air above the tropopause?

- a. The temperature stops falling and stays constant
- b. The temperature continues to fall at the same rate
- c. The temperature begins to rise with height

306. A performance calculation is made for 3000 m in the standard atmosphere, where the pressure is 701.09 hPa. Which air temperature belongs to that altitude?

- a. + 2.0 °C
- b. − 17.5 °C
- c. − 4.5 °C

307. An airfield lies 1000 m below mean sea level. Which standard pressure and temperature apply there?

- a. 1139.29 hPa and 21.5 °C
- b. 1013.25 hPa and 15.0 °C
- c. 898.75 hPa and 8.5 °C

308. Standard conditions give 15 °C at mean sea level and a lapse of 0.65 °C per 100 m. Which temperature results at 3000 m, and does it agree with the ISA table?

- a. − 4.5 °C, which is colder than the table
- b. − 4.5 °C, which agrees with the table
- c. + 4.5 °C, which agrees with the table

309. Standard density is 1.2250 kg/m³ at sea level and 0.3692 kg/m³ at the tropopause. Which density ratio does an aircraft meet at the tropopause?

- a. About 60 % of the sea level value
- b. About 30 % of the sea level value
- c. About 3 % of the sea level value

Altitude, altimeter settings and RVSM

310. During the climb-out the crew change the altimeter setting from QNH to 1013.2 hPa. Which point has the aircraft reached?

- a. The transition level
- b. The elevation reference
- c. The transition altitude

- 311.** On descent for landing the crew change the altimeter setting from STD back to QNH. Which point has the aircraft reached?
- The transition level
 - The pressure altitude
 - The transition altitude
- 312.** An airfield has an elevation of 2712 ft and an aircraft above it indicates an altitude of 5319 ft on QNH. Which height above the airfield results?
- 5319 ft
 - 8031 ft
 - 2607 ft
- 313.** An aircraft parked on an airfield reads exactly zero on its barometric altimeter. Which setting is in use?
- QFE
 - QFF
 - QNH
- 314.** A transition altitude of 5000 ft MSL is published for an airfield. Which is the lowest transition level that may be specified with it?
- One at least 1000 ft below, so about 4000 ft
 - One at the same height, so about 5000 ft
 - One at least 1000 ft above, so about 6000 ft
- 315.** An aircraft flies over an airfield of 2712 ft elevation with QFE set on the altimeter. Which value does the altimeter show?
- The elevation of the airfield
 - The height above the airfield
 - The altitude above mean sea level
- 316.** An aircraft cruises at 4000 ft above mean sea level in an area with a transition altitude of 5000 ft. Which altimeter setting must be used?
- QNH
 - QFF
 - QNE
- 317.** An aircraft parked on an airfield of 2712 ft elevation shows exactly 2712 ft on its altimeter. Which setting is in use?
- QNH
 - QFE
 - QNE
- 318.** The same pressure change of 1 hPa is worth 30 ft at sea level but 120 ft at FL 360. Which explanation applies?
- Air density and pressure fall with altitude, so a given pressure step spans more altitude
 - Air density and pressure rise with altitude, so a given pressure step spans more altitude
 - Air temperature falls with altitude, so the altimeter capsule expands more
- 319.** A dent is found in the skin close to a static port of an RVSM approved aircraft. Which consequence must be expected?
- A considerable influence on the radio altimeter indication
 - A considerable influence on the altitude indication
 - A considerable influence on the mach number only
- 320.** Two IFR aircraft cruise on opposite tracks at FL 330 in RVSM airspace today. Which vertical separation applies, and which applied before 1997?
- 1000 ft today and 2000 ft before 1997
 - 1000 ft today and 1000 ft before 1997
 - 2000 ft today and 1000 ft before 1997
- 321.** A static system leak gives an error of 4 hPa while the aircraft is cruising at FL 360. Which altitude error results?
- About 30 ft
 - About 120 ft
 - About 480 ft
- 322.** An altimeter fails in cruise and the aircraft loses its RVSM capability. Which action is required?
- A change of the altimeter setting to QNH
 - A quick exit of RVSM airspace
 - A descent to the transition level
- 323.** An aircraft holds RVSM Airworthiness Approval while its altitude monitoring programme and test flight are still outstanding. May it use RVSM airspace?
- Yes, the Airworthiness Approval alone is sufficient
 - Yes, provided two independent altimeters are fitted
 - No, RVSM Operational Approval is still needed

Airspeed

324. An air data computer shows 293 knots CAS at 30,000 ft and the compressibility error is 15 knots. Which equivalent airspeed results?

- a. 308 knots
- b. 263 knots
- c. 278 knots

325. Why is the equivalent airspeed lower than the calibrated airspeed at high speed and altitude?

- a. Because compression of the air lowers the static pressure, so the indication is too low
- b. Because reduced density lowers the total pressure, so the indication is too high
- c. Because compression of the air raises the total pressure, so the indication is too high

326. At 30,000 ft the TAS in the example is far above the EAS. Which quantity explains the difference between the two?

- a. The static source error of the instrument
- b. The air density at the flight altitude
- c. The compressibility of the air at high speed

327. Two aircraft fly at the same calibrated airspeed, one at 20,000 ft and one at 40,000 ft. Which one has the larger compressibility error?

- a. Both have the same error
- b. The aircraft at 20,000 ft
- c. The aircraft at 40,000 ft

Barometers, capsules and instrument errors

328. The pressure supplied to a Bourdon tube transducer is doubled. Which effect does this have on the output signal?

- a. The signal changes in proportion because the tip moves further
- b. The signal stays the same because the tube is sealed at the tip
- c. The signal changes in steps because the sensor counts tube turns

329. An altimeter needle must turn 51 times over its full range. Which demand does this place on the instrument?

- a. The highest demands on electrical power and signal bandwidth
- b. The highest demands on material and fine mechanical precision
- c. The highest demands on case sealing and static line length

330. Which measures are used to characterize the properties of a capsule for its intended use

- a. Special alloys, natural ageing and particular residual pressures
- b. Special coatings, artificial ageing and particular bearing materials
- c. Special alloys, artificial ageing and particular waveforms

331. Which parts make temperature compensation demanding over the range -70 to +50 degrees C

- a. The capsule mounting and the flexible tubes in the static system
- b. The pinion bearings and the quick release connectors of the instrument
- c. The capsule mounting and the variable steering levers in the indicator mechanism

332. An altimeter is held at one steady pressure on the test bench for a long time and the indication slowly drifts. Which error influence explains this?

- a. The hysteresis of the capsule
- b. The elastic aftereffect of the capsule
- c. The backlash of the pinion gearing

333. An altimeter reads slightly differently at the same test pressure depending on whether the pressure was raised or lowered to reach it. Which error influence is shown?

- a. Elastic aftereffect of the diaphragm capsule
- b. Hysteresis of the diaphragm capsule
- c. Friction in the transfer mechanism

334. Through which parts must the smallest pressure change act to deflect the needle exactly

- a. From the capsule via levers, shaft and cogwheel
- b. From the capsule via bellows, resonator and coil
- c. From the diaphragm via strain gauges, bridge and amplifier

Pressure transducers

335. A force compensated pressure cell is fitted instead of a plain pressure cell. Which two advantages are gained?

- a. It is sensitive to pressure changes and has small instrument errors
- b. It is insensitive to pressure changes and needs no separate pressure cells
- c. It is insensitive to pressure changes and has small instrument errors

- 336.** Which of these pressure sensing types needs no different pressure cells for static and pitot?
- The plain pressure cell type
 - The force compensated pressure cell
 - The vibration pressure transducer
- 337.** An air data computer needs one unit that works for both static and pitot pressure and gives a digital output. Which of these meets that?
- The vibration pressure transducer
 - The force compensated pressure cell
 - The plain pressure cell with resolver
- 338.** A vibrating transducer is cut open. Where is the pressure that is being measured found?
- In the pressure chamber inside the vibrating cylinder
 - In the reference chamber outside the vibrating cylinder
 - In the pressure chamber inside the center body
- 339.** What is done with the signal from the pickup coil
- It is amplified and used as output, and fed back to the drive coil
 - It is amplified and used as output, and fed back to the reference chamber
 - It is digitized and used as reference, and fed back to the drive coil
- 340.** Why is the pickup coil signal fed back to the drive coil of a vibrating transducer?
- To keep the output frequency at a constant value
 - To keep the reference chamber at a constant vacuum
 - To keep the cylinder vibrating with its natural frequency
- 341.** The pressure supplied to a vibrating transducer changes. Which output quantity changes as a result?
- The frequency of the output signal
 - The reference pressure in the chamber
 - The amplitude of the output signal
- 342.** With U1 at 8 volts and U2 at 2 volts, which output voltage does the bridge deliver?
- 4 volts
 - 10 volts
 - 6 volts
- 343.** Pressure is applied to a piezoresistive transducer. How do the two bridge voltages behave?
- U1 and U2 both increase
 - U1 increases while U2 decreases
 - U1 decreases while U2 increases
- 344.** A strain gauge wire is made thicker while its length stays the same. How does its resistance change?
- It increases with increasing area
 - It decreases with increasing area
 - It is independent of the area
- 345.** A strain gauge on the bent side of the diaphragm is stretched. Why does its resistance rise?
- Because the wire becomes longer and its cross section smaller
 - Because the material constant of the wire rises with the strain
 - Because the wire becomes shorter and its cross section larger

Answer key

No.	Ans	Level	LTT page	Why
1	c	L1	2	The text splits classical cockpit instruments into flight monitoring and aircraft monitoring.
2	a	L1	2	The classification uses exactly two subgroups.
3	a	L1	2	The ADI is listed among the flight monitoring instruments.
4	c	L1	2	The machmeter is listed among the flight monitoring instruments.
5	c	L1	2	RPM indicators are engine monitoring devices, which form part of aircraft monitoring.
6	b	L1	2	The magnetic compass is listed among the flight monitoring instruments.
7	c	L1	2	The turn and slip indicator appears in the flight monitoring list; the others are aircraft monitoring devices.
8	c	L1	2	Position indicators and RPM indicators are both aircraft monitoring devices.
9	a	L1	2	Airframe monitoring devices are given as position indicators and pressure indicators.
10	b	L1	2	The engine monitoring devices listed are RPM, exhaust gas temperature and fuel indicators.
11	c	L1	2	EIS is given as Electronic Indicating System.
12	c	L1	2	EFIS is given as Electronic Flight Instrument System.
13	b	L1	2	Flight monitoring instruments are assigned to the EFIS.
14	c	L1	2	EICAS is given as Engine Indicating and Crew Alerting System.
15	a	L1	2	ECAM is given as Electronic Centralized Aircraft Monitoring.
16	c	L1	2	CMS is given as Central Maintenance System.
17	b	L1	2	The standby devices listed are altimeter, airspeed indicator, horizon and compass.
18	a	L1	2	Four standby devices are listed: altimeter, airspeed indicator, horizon and compass.
19	b	L1	2	Earth's magnetism is one of the listed values whose change causes instruments to operate.
20	b	L1	2	Air pressure, temperature and acceleration all appear in the list of values.
21	b	L1	2	The list of values names gyro stability, together with inertia and acceleration.
22	c	L1	8	The manual states the instrumentation is duplicated so that each pilot has his own set.
23	c	L1	14	Those three parameters are the ones the Basic T set is defined to present.
24	c	L1	14	The variometer is listed as the rate of climb/dive instrument of the set.
25	b	L1	14	The Basic Six list gives the airspeed indicator in knots and the altimeter in feet.
26	c	L1	14	The set lists the altimeter in feet.
27	b	L1	14	The directional gyro is named as the compass of the Basic Six.
28	b	L1	14	The list gives the artificial horizon as the aircraft attitude instrument.
29	b	L1	14	The set is defined as six essential instruments, hence the name Basic Six.
30	a	L1	16	The four instruments arranged in the T shape are the airspeed indicator, attitude indicator, altimeter and compass; turn coordination and vertical speed sit below them.
31	b	L1	16	The Basic T places the attitude indicator centrally at the top of the group.
32	b	L1	16	The altimeter occupies the position to the right of the attitude indicator.
33	a	L1	16	The airspeed indicator takes the left arm of the T.
34	b	L1	16	The compass forms the stem of the T below the attitude indicator.
35	a	L1	16	Its usual mounting position is between the windcreens, clear of the main panel.
36	c	L1	16	The upper row is given as airspeed, artificial horizon and altimeter.
37	a	L1	16	Those three are listed as the lower row of the Basic T.
38	c	L1	16	ADI is the Attitude Director Indicator.
39	c	L1	16	The ADI combines attitude with flight director guidance.
40	c	L1	16	HSI is the Horizontal Situation Indicator.
41	b	L1	16	PFD stands for Primary Flight Display.
42	b	L1	16	Vertical speed is the fifth Basic T parameter listed on the PFD.
43	b	L1	17	The HSI occupies the position under the ADI in the Basic T group.
44	b	L1	17	Lateral deviation is shown along the bottom, below the attitude area.
45	c	L1	17	Vertical deviation occupies the right-hand strip, opposite the airspeed scale.
46	a	L1	18	Simplification and centralisation are the two words the manual uses for the trend.
47	a	L1	19	Touch, viewing direction control and voice are the next stage shown for control input.
48	b	L1	19	Navigation is grouped onto the navigation display.
49	b	L1	20	Nitrogen makes up 78% of dry clean air at the surface.

50	c	L1	20	Oxygen makes up 21 % of dry clean air at the surface.
51	a	L1	20	The remaining one per cent is the noble gas group named in the text.
52	c	L1	20	The troposphere is the layer in contact with the earth's surface.
53	b	L1	20	Weather is contained in the troposphere.
54	c	L1	20	At the poles the troposphere reaches only about 8.5 km.
55	b	L1	21	Above the troposphere and its tropopause boundary comes the stratosphere.
56	b	L1	21	The tropopause caps the troposphere.
57	b	L1	21	The stratosphere extends to about 50 km, where the stratopause lies.
58	a	L1	21	The ozone layer sits in the stratosphere, which is why that region is also called the ozonosphere.
59	a	L1	21	The exosphere is the highest layer shown.
60	c	L1	22	Flight monitoring is built on the two pressures, static and dynamic.
61	b	L1	22	Dynamic pressure is the ram pressure produced by the aircraft's motion.
62	c	L1	22	Static pressure is the atmospheric pressure surrounding the aircraft.
63	a	L1	22	Those four come from the pressures alone; TAS and SAT need temperature in addition.
64	b	L1	22	ISA sea level density is 1.225 kg per cubic metre.
65	b	L1	22	ISA mean sea level is 15 °C at 1013 mbar.
66	a	L1	22	Barometric altitude comes from static pressure only.
67	a	L1	23	Everest is still well below the tropopause, inside the troposphere.
68	a	L1	24	1013.25 hPa is the same pressure as 29.92 inches of mercury.
69	b	L1	24	The imperial figure for ISA sea level density is 0.002378 slug per cubic foot.
70	c	L1	24	15 °C corresponds to 59 °F in the ISA table.
71	b	L1	24	The ISA assumes completely dry air.
72	b	L1	24	The metric ISA lapse rate is 0.65 °C for every 100 m up to the tropopause.
73	b	L1	24	ISA gives –56.5 °C at the 11000 m tropopause.
74	a	L1	24	The standard pressure at the tropopause is 226.32 hPa.
75	b	L1	24	11000 m is listed as 36089 ft.
76	a	L1	24	At the tropopause the standard density is 0.3692 kg per cubic metre.
77	b	L1	25	760 mm Hg is the same sea level pressure as 1013.25 mb.
78	b	L1	25	ISA sea level pressure is 14.696 psi.
79	a	L1	25	The chart places the tropopause at 11 km.
80	a	L1	28	All air data is brought together on the Primary Flight Display.
81	a	L1	28	HUD is short for Head Up Display.
82	b	L1	29	Airspeed occupies the left-hand tape of the PFD.
83	c	L1	30	The reference is entered on the altimeter's pressure scale, and the indication follows from it.
84	b	L1	30	Elevation refers to something on the ground measured from mean sea level.
85	b	L1	30	Elevation carries the setting code QFF.
86	c	L1	30	Height is measured from the ground below the aircraft.
87	b	L1	30	Height above ground is normally taken from the low range radio altimeter.
88	c	L1	30	Barometric altitude above mean sea level is flown on QNH.
89	c	L1	30	QNH is the actual pressure reduced to mean sea level.
90	a	L1	30	The standard setting is coded QNE.
91	c	L1	30	Standard setting is 1013.2 hPa.
92	a	L1	30	With 1013.2 hPa set, the indication is pressure altitude.
93	c	L1	30	Flight levels are counted from the standard 1013.25 hPa pressure surface.
94	a	L1	30	At least 1000 ft of margin is required.
95	b	L1	32	Reduced vertical separation minimum halves the vertical spacing used up high.
96	c	L1	32	Below FL 285 the separation has always been 1000 ft.
97	c	L1	32	Above FL 285 the spacing used to be doubled to 2000 ft.
98	c	L1	32	At sea level 1 hPa is worth about 30 ft.
99	a	L1	32	High up, the same 1 hPa covers 120 ft.
100	c	L1	32	RVSM airspace starts above FL 290 and uses 1000 ft spacing.
101	b	L1	32	MNPS (Minimum Navigation Performance Specifications) is the specification named for RVSM approval.

102	c	L1	32	The autopilot altitude hold mode is on the list of RVSM-sensitive items.
103	b	L1	32	Two independent altimeters are required.
104	c	L1	32	Europe, America and the Pacific are the regions named.
105	b	L1	34	Training covers both flight crew and maintenance staff.
106	a	L1	34	The pitot-static sensors are replaced every 9,600 hours.
107	b	L1	34	SSEC is the correction applied for static source error.
108	b	L1	34	MNPS approval is added to the existing long range navigation approval.
109	a	L1	36	The text defines IAS as indicated airspeed.
110	b	L1	36	Glass cockpits with an ADC display CAS.
111	c	L1	36	The example gives CAS as 7 knots below IAS.
112	b	L1	37	Ptot and Pstat are the two inputs shown to the ADC.
113	c	L1	37	The example chain gives 300 kts IAS and 293 kts CAS.
114	a	L1	38	The worked example gives a 15 knot compressibility error at that condition.
115	a	L1	38	The manual states TAS is required for navigation.
116	c	L1	38	The example gives 470 knots TAS for 278 knots EAS at 30,000 ft.
117	c	L1	40	The tube stands vertically and is closed at the top.
118	b	L1	40	The manual gives 760 mm as the normal mercury column height.
119	a	L1	40	The aneroid element is a thin metal capsule.
120	a	L1	40	The capsule keeps a residual pressure of about 5 mbar.
121	c	L1	40	The manual gives 5 mbar as 5 hPa or 500 Pa.
122	b	L1	40	The manual states up to eight capsules.
123	b	L1	42	The Bourdon tube arrangement turns pressure into a proportional electrical output.
124	c	L1	42	The tube has an oval cross section in the unpressurised state.
125	c	L1	42	The manual names the synchro transmitter as the usual position sensor.
126	b	L1	46	The required range is -70 °C to +50 °C.
127	b	L1	46	The manual gives 51 needle revolutions over that range.
128	c	L1	47	On the panel the variometer is the instantaneous vertical speed indicator.
129	c	L1	48	The supply is by piping and flexible tubes with screw or quick release connectors.
130	b	L1	48	The capsule stroke is only 4 to 5 mm.
131	c	L1	50	Transducers turn pressure into electrical signals.
132	b	L1	50	The manual names air data computers as an application.
133	c	L1	50	The newest transducers give a digital output.
134	a	L1	50	The reference is either a vacuum or 1013 mbar.
135	b	L1	51	The pressure cell with resolver gives an analog output.
136	a	L1	51	The force compensated cell uses a torsion bar and an E-pickup.
137	c	L1	51	The vibration type converts with an oscillator and a resonator.
138	b	L1	51	The piezoresistive type converts with a resistive bridge.
139	a	L1	52	The manual calls the vibrating transducer the most commonly used type.
140	c	L1	52	The three named parts are base plate, center body and outer case.
141	c	L1	52	The reference chamber usually holds a vacuum.
142	c	L1	54	The chamber is made of silicon.
143	b	L1	54	The structure measures only a few millimetres.
144	a	L1	54	The construction uses silicon with Pyrex glass.
145	b	L1	54	UIN is the supply and UOUT the measuring voltage.
146	c	L1	55	The strain gauges are wired as a Wheatstone bridge.
147	c	L1	56	Normally four strain gauges are used.
148	c	L1	56	The 10 volt supply divides equally into 5 volts on each side.
149	a	L1	56	In the example U2 falls to 2 volts while U1 rises to 8 volts.
150	a	L1	57	The bridge is supplied with 10 volts in the example.
151	c	L2	2	Exhaust gas temperature indicators are named as engine monitoring devices under aircraft monitoring.
152	c	L2	2	Aircraft monitoring covers airframe monitoring devices and engine monitoring devices.
153	b	L2	2	Pressure indicators are listed as airframe monitoring devices under aircraft monitoring.

154	b	L2	2	Altimeter and airspeed indicator are flight monitoring instruments, and these are assigned to the EFIS.
155	c	L2	2	The text states that aircraft monitoring instruments have different names depending on aircraft type.
156	b	L2	2	EICAS and ECAM are the names given to the aircraft monitoring instruments.
157	c	L2	2	RPM is an engine monitoring value, so it belongs to aircraft monitoring, named EICAS or ECAM.
158	b	L2	2	The CMS is used for aircraft monitoring because it delivers information to maintenance personnel.
159	b	L2	2	Flight monitoring goes to the EFIS and aircraft monitoring is named EICAS or ECAM.
160	a	L2	2	The standby devices are altimeter, airspeed indicator, horizon and compass; the machmeter is not among them.
161	b	L2	2	Fluid pressure is listed separately from air pressure as a value that operates instruments.
162	b	L2	2	Earth's magnetism is the listed value a magnetic compass responds to.
163	a	L2	2	Air pressure is the listed value on which altimeter and airspeed indications depend.
164	b	L2	8	Duplication per pilot station is exactly what the general rule for flight monitoring instruments describes.
165	a	L2	14	The identical panel layout removes most of the retraining needed for blind flying on a new type.
166	c	L2	16	The HSI is the instrument that combines compass and radio navigation information.
167	b	L2	16	The ADI carries attitude and flight director, the HSI carries compass and radio navigation.
168	b	L2	16	The PFD keeps the Basic T arrangement rather than abandoning it.
169	c	L2	17	Attitude with guidance occupies the middle of the PFD, in place of the ADI.
170	a	L2	19	The individual operating devices are drawn together into integrated central control units.
171	a	L2	19	Attitude is grouped onto the PFD and, where fitted, the HUD.
172	a	L2	19	Engine information goes to the engine/warning display and systems to the system display.
173	a	L2	20	The troposphere is deepest at the equator, about 16.8 km, against about 8.5 km at the poles.
174	a	L2	20	The troposphere lapse rate is 0.65 °C per 100 m of climb.
175	c	L2	21	Above the mesopause the temperature rises again through the thermosphere.
176	c	L2	22	All three listed values fall with increasing altitude.
177	c	L2	22	Temperature is the corrective value that makes TAS and SAT available.
178	b	L2	22	Calibration is referred to the ISA mean sea level values.
179	c	L2	22	The ISA is the ICAO standard atmosphere.
180	a	L2	23	The climb takes the aircraft into air of lower pressure, lower temperature and lower density.
181	b	L2	24	The ISA exists so performance figures can be compared on a common basis.
182	b	L2	25	Sea level density is the reference, so its ratio is 100 %.
183	a	L2	26	Either the instrument does the whole job or a separate air data calculator does the conversion.
184	a	L2	26	The calculator turns pressure into an electrical signal that drives electrical instruments.
185	b	L2	26	The manual attributes the differences to manufacturer and technology level.
186	b	L2	28	The HUD projects the PFD information into the pilot's forward view.
187	a	L2	28	Only the head up display lets him read the data while looking outside.
188	a	L2	30	With QFE set the reading starts from the airfield, so it is zero on the ground.
189	b	L2	30	QFE gives height above the airfield, valid while the pressure stays as set.
190	a	L2	30	QNH refers the altimeter to mean sea level, so on the ground it reads the airfield elevation.
191	c	L2	30	QNH belongs to the low level phases, below the transition altitude or level.
192	a	L2	30	The transition altitude quoted here is 5000 ft MSL.
193	c	L2	30	A common reference means all altimeters err alike, which keeps the vertical spacing valid.
194	a	L2	30	The transition altitude is the climb-out point for changing to the standard setting.
195	b	L2	30	The transition level is the descent point for changing back to QNH.
196	b	L2	31	1013.25 hPa is the standard ISA sea level value on that scale.
197	a	L2	31	A flight level is always counted from the 1013.25 hPa surface.
198	a	L2	32	More usable levels mean more aircraft fit into the same volume of airspace.
199	a	L2	32	Altimeters become less precise with height, so a bigger buffer was needed.
200	a	L2	32	SSE arises from where the static inlet sits in the airflow over the skin.
201	a	L2	32	The area around the sensors has to stay clean and clear so the sensed pressure is undisturbed.
202	b	L2	32	The device warns the crew that the selected altitude is being left.
203	a	L2	32	RVSM depends on the autopilot holding the selected altitude accurately.
204	a	L2	32	The transponder must report altitude taken from the altitude measurement unit.
205	b	L2	32	More available levels give better routings, saving time and fuel.

206	c	L2	33	Full implementation stretches the band up to FL 420.
207	a	L2	33	Above FL290 in the final implementation both the MNPS and the RVSM requirements apply.
208	c	L2	34	Approval is per aircraft and is applied for by the operator.
209	c	L2	34	Both the service bulletin modifications and the altitude monitoring programme are required.
210	a	L2	34	Those three are the surface waviness causes listed.
211	c	L2	34	The training list names the part numbers of the air data computer.
212	b	L2	34	Airworthiness approval comes first and opens the way to the test flight.
213	c	L2	34	Operational approval is the last step and permits use of RVSM airspace.
214	a	L2	34	Both the operation and the maintenance manuals are revised.
215	b	L2	36	IAS follows dynamic pressure, which is total pressure minus static pressure.
216	b	L2	36	The bellows instrument is only true at mean sea level in standard conditions.
217	a	L2	36	CAS is IAS corrected for instrument error and static source error.
218	a	L2	36	A bellows instrument with no computer correction shows IAS.
219	a	L2	37	The ADC receives total and static pressure and applies the corrections that turn IAS into CAS.
220	a	L2	38	The formula assumes an incompressible medium.
221	b	L2	38	Compression of the air at high speed raises the total pressure, which the instrument reads as extra speed.
222	c	L2	38	The error grows with both airspeed and altitude.
223	c	L2	38	EAS is what is left when the compressibility error is taken out of CAS.
224	a	L2	38	EAS is a design and calculation value, not a cockpit indication.
225	c	L2	38	TAS is the aircraft speed relative to the surrounding air mass.
226	c	L2	39	The summary chain runs IAS to CAS to EAS to TAS.
227	b	L2	39	The last step of the chain is the density calculation.
228	a	L2	39	CAS becomes EAS when the compressibility error is corrected.
229	c	L2	39	The 15 knot compressibility error is taken off the calibrated airspeed of 293 knots to give the equivalent airspeed of 278 knots.
230	c	L2	40	The weight of the fluid draws it down and leaves a negative pressure above it.
231	b	L2	40	Ambient static pressure on the container balances the column and holds it at a height.
232	c	L2	40	Both fluid and tube expand with temperature, so the reading must be corrected.
233	a	L2	40	Exact results need the raw reading reduced to standard conditions.
234	a	L2	40	The small residual pressure offsets temperature-driven changes in capsule elasticity.
235	c	L2	40	Stacking capsules is done to make the measurement more reliable.
236	b	L2	40	Rising pressure squeezes the capsule, so it contracts.
237	c	L2	40	Falling pressure lets the capsule expand.
238	c	L2	40	The capsule stroke drives the needle by mechanical linkage.
239	a	L2	41	The aircraft aneroid altimeter senses static pressure acting on an evacuated capsule.
240	b	L2	41	Ambient pressure pushes on the open container and a negative pressure sits above the mercury column.
241	c	L2	41	The aneroid altimeter carries a temperature compensation in addition to the capsule and the indicating linkage.
242	c	L2	42	Only the curved oval tube and a position sensor at its tip are required.
243	c	L2	42	The sensor senses tip position, not pressure directly.
244	b	L2	44	The pressure acts equally over the whole tube.
245	a	L2	44	Pressure rounds the oval section into a circular one.
246	b	L2	44	Rounding the section straightens the curved tube and carries the tip to the right.
247	c	L2	44	Tip travel is proportional to the applied pressure.
248	c	L2	46	The requirement list includes insensitivity to acceleration.
249	b	L2	46	The requirements are laid down in law, so the design freedom is small.
250	a	L2	46	Direct pressure sensor instruments survive only as back-up instruments.
251	a	L2	46	A stack of thin metal diaphragm capsules is called a bellows.
252	b	L2	47	All four of these are pressure sensing instruments shown on the panel.
253	b	L2	48	The mechanism amplifies the tiny stroke into needle rotation.
254	b	L2	48	Friction in the mechanism corrupts the reading.
255	b	L2	48	Good bearings and clean mechanical construction minimise friction.
256	c	L2	48	Vibration keeps the altimeter mechanism from sticking, helping against friction.
257	b	L2	48	A one-sided spring tension takes up the gear backlash.

258	a	L2	48	Hysteresis is the difference between the rising and falling pressure characteristics.
259	a	L2	48	Long-held constant pressure makes the capsule yield slowly.
260	c	L2	48	The stroke lags in time, which is the elastic aftereffect.
261	b	L2	48	The manual states these errors act on all aneroid instruments.
262	a	L2	49	Temperature compensation in the mechanical altimeter is done with a bimetal element.
263	a	L2	49	Both capsule errors are shown as the diaphragm stroke plotted against the applied pressure.
264	b	L2	49	The pinion on the pointer axis turns the pointer.
265	b	L2	50	The signals go to the computer, which makes the required outputs.
266	a	L2	50	Early transducers used metal capsules working electromotively on the follow-up principle.
267	a	L2	50	Modern transducer chambers are of semiconductive material.
268	b	L2	50	The built-in memory holds the error characteristics of that unit.
269	a	L2	50	The metal diaphragm of the pressure cell works against a vacuum.
270	a	L2	51	The plain pressure cell is sensitive to pressure changes and mechanically complex.
271	b	L2	51	The vibration type compensates its errors electrically.
272	a	L2	51	The vibration type is described as electronically complex.
273	c	L2	51	Both the vibration and the piezoresistive types give digital outputs.
274	a	L2	51	The two pressure cell types give analog outputs.
275	a	L2	52	Its output is a frequency that varies with the measured pressure.
276	c	L2	52	The outer case is the outer wall of the reference chamber.
277	c	L2	52	The vibrating cylinder separates the reference chamber from the measured pressure.
278	b	L2	52	The drive coil field sets the cylinder vibrating at its natural frequency.
279	c	L2	52	Both the body and the conditions around it set the natural frequency.
280	a	L2	52	The pickup coil senses how fast the cylinder is vibrating.
281	c	L2	53	The output of the vibrating transducer is the frequency f .
282	a	L2	53	The manual compares the cylinder to violin strings and tuning forks.
283	a	L2	54	Denting the lid deforms the surface resistors and changes their resistance.
284	c	L2	54	The resistance change is the measure of pressure.
285	c	L2	54	The legend gives R1 as stretched and R2 as compressed.
286	a	L2	54	The output voltage rises in proportion to the applied pressure.
287	c	L2	54	The analogue bridge voltage is digitized for the computer.
288	a	L2	55	The diaphragm separates the input pressure chamber from the reference pressure chamber.
289	b	L2	55	A straight rising characteristic means the output voltage is proportional to the applied pressure.
290	b	L2	56	Two gauges sit on the front and two on the back.
291	a	L2	56	With no pressure the plate stays straight, all resistances are equal and the bridge balances.
292	a	L2	56	Equal voltages on both sides give zero output.
293	b	L2	56	Bending lowers the front resistance and raises the back one.
294	b	L2	56	Unequal resistances unbalance the bridge and so the output.
295	b	L2	56	Resistance depends on the material constant and on the wire dimensions.
296	a	L2	56	Only the dimensions change with bending, not the material constant.
297	c	L2	56	Longer wire gives more resistance.
298	a	L2	57	Figure 35 labels the two faces of the diaphragm R1 plus R4 on one side and R2 plus R3 on the other.
299	a	L2	57	The formula is R equals the material constant times length over area.
300	a	L2	57	With no pressure the diaphragm and its gauges stay flat and unstrained.
301	c	L3	2	The CMS is the aircraft monitoring system whose information goes to maintenance personnel.
302	a	L3	2	Flight monitoring is assigned to the EFIS, while only the aircraft monitoring system takes different names by aircraft type.
303	c	L3	2	The standby set contains an airspeed indicator and a horizon, but no machmeter.
304	c	L3	20	Twenty steps of 100 m at 0.65 °C each give a fall of 13 °C.
305	a	L3	24	Identical temperatures at both levels show the lapse rate has stopped at the tropopause.
306	c	L3	24	The 3000 m row of the ISA table gives -4.5 °C.
307	a	L3	24	Below sea level the standard values rise, to 1139.29 hPa and 21.5 °C at -1000 m.
308	b	L3	24	Thirty steps of 0.65 °C take 15 °C down to -4.5 °C, exactly the table value.
309	b	L3	24	0.3692 divided by 1.2250 is close to three tenths, so about 30 % of sea level density.

310	c	L3	30	Changing from QNH to standard on the climb defines the transition altitude.
311	a	L3	30	Changing back to QNH on descent happens at the transition level.
312	c	L3	30	Height above the airfield is the altitude minus the airfield elevation, $5319 - 2712 = 2607$ ft.
313	a	L3	30	A zero reading on the ground is the QFE setting.
314	c	L3	30	The transition level must lie at least 1000 ft above, giving roughly 6000 ft.
315	b	L3	30	QFE refers the altimeter to the airfield, so it reads height above it, not altitude above sea level.
316	a	L3	30	Standard setting only applies above the transition altitude, so below it the altimeter stays on QNH.
317	a	L3	31	Only QNH makes the altimeter read the airfield elevation on the ground.
318	a	L3	32	Thin air up high means pressure changes slowly with height, so one hPa covers far more feet.
319	b	L3	32	Skin damage near the static source disturbs the static pressure and so the altitude reading.
320	a	L3	32	RVSM halved the old 2000 ft spacing above FL 285 back to 1000 ft.
321	c	L3	32	Each hPa is worth 120 ft at FL 360, so four hPa give about 480 ft.
322	b	L3	34	Losing RVSM capability means the aircraft must leave RVSM airspace quickly.
323	c	L3	34	Use of RVSM airspace only follows the Operational Approval.
324	c	L3	38	The compressibility error is subtracted from CAS, giving $293 - 15 = 278$ knots.
325	c	L3	38	The extra total pressure from compression makes the reading too high, so the correction reduces the speed.
326	b	L3	38	The step from EAS to TAS is a density calculation, and low density at altitude makes TAS much higher.
327	c	L3	39	The error grows with altitude, so the higher aircraft has the larger error.
328	a	L3	44	Tip travel is proportional to pressure and the sensor output follows tip position.
329	b	L3	46	Such fine resolution needs the best material and fine mechanics.
330	c	L3	48	The three listed measures are special alloys, artificial ageing and particular waveforms.
331	c	L3	48	Compensation is hard because of the capsule mounting and the variable steering levers.
332	b	L3	48	Slow drift under a constant held pressure is the elastic aftereffect, not hysteresis.
333	b	L3	48	A difference between the rising and falling approach to the same pressure is hysteresis.
334	a	L3	48	The mechanical chain runs capsule, levers, shaft, cogwheel, needle.
335	c	L3	51	The force compensated cell is insensitive to pressure changes and keeps instrument errors small.
336	c	L3	51	The table gives 'no different pressure cells for static and pitot' for the vibration type; both pressure cell types need separate cells.
337	a	L3	51	Of the three offered, only the vibration type gives a digital output and needs no separate cells for static and pitot.
338	a	L3	52	The measured pressure sits in the chamber inside the cylinder.
339	a	L3	52	The pickup signal serves both as the amplified output and as the feedback to the drive coil.
340	c	L3	52	The feedback loop sustains vibration at the natural frequency.
341	a	L3	52	The measured pressure alters the cylinder natural frequency, so the output frequency changes.
342	c	L3	56	The output is the difference between U1 and U2, that is 8 minus 2 volts.
343	b	L3	56	The two halves of the bridge move in opposite directions when the plate bends.
344	b	L3	56	A larger cross section lowers the resistance.
345	a	L3	56	Stretching lengthens the wire and thins it, and both changes raise the resistance.